

FORECASTING U.S. INCOME INEQUALITY WITH AUTOREGRESSIVE MODELS: A LAYERED APPROACH USING THE GINI COEFFICIENT

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ABSTRACT. Income inequality remains a central concern in economic policy, yet forecasting its trajectory poses significant statistical challenges due to the persistent, slow-moving nature of inequality measures. This project develops a layered modeling framework to forecast the U.S. Gini coefficient over a 30-year out-of-sample window (1994–2023) using annual data from 1963 onward. Each successive layer adds complexity: Layer 1 fits a deterministic linear trend as a baseline, Layer 2 introduces an autoregressive model to capture year-to-year persistence, and Layer 3 extends it with moving-average, differencing, and exogenous macroeconomic components (ARX, ARMA, ARIMA, ARIMAX). All layers are evaluated through rolling forecasts that simulate real-time prediction; the linear models are estimated by ordinary least squares and the likelihood-based models by numerical optimization, implemented in Python with NumPy and SciPy. We compare the layers across standard error metrics, interpret the estimated parameters in economic terms, and discuss where the models succeed and where they struggle.

1. INTRODUCTION

Income inequality is one of the most studied and debated topics in economics and public policy. During the past six decades, the United States has experienced a pronounced upward trend in income inequality, a shift that has reshaped debates about taxation, labor markets, and social mobility. Quantifying this trend and understanding its dynamics are essential first steps for anyone seeking to analyze or forecast inequality over time.

The Gini coefficient is the most widely used single-number summary of income inequality in a population. It ranges from 0 (perfect equality) to 1 (perfect inequality) and is available for the United States on an annual basis since 1963 through Our World in Data, drawing on the World Bank’s Poverty and Inequality Platform [1]. Despite its simplicity, the Gini coefficient captures the broad sweep of distributional change, and its availability as a long, consistent time series makes it a natural candidate for statistical modeling and forecasting.

The central question motivating this project is: *Can statistical time-series models forecast the long-term evolution of U.S. income inequality?* Recent studies have applied ARIMA and regression models to forecast Gini coefficients in the U.S. [3] and across multiple countries [7], while the same Box–Jenkins methodology has been widely used for GDP [4, 6] and unemployment forecasting [5]. Rather than attempting to answer this question with a single model, we adopt a *layered* approach in which each successive layer adds complexity and explanatory power, with each new layer designed to fix a limitation of the previous one.

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Layer	Model	What it adds
1	Linear trend	Simple straight-line baseline
2	AR(1)	Uses last year's value to predict next year
3	AR families	Adjusts for trends, past mistakes, and economic data

The data consist of 61 annual U.S. Gini coefficient values from 1963 to 2023. We split the sample into a training period (1963–1993, 31 years) and a test period (1994–2023, 30 years). All models are estimated using ordinary least squares or maximum likelihood estimation, and forecasts are generated via an expanding-window rolling procedure that avoids look-ahead bias. Forecast quality is measured by mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

The remainder of this paper is organized as follows. Section 2 provides the mathematical background needed to understand the modeling framework. Section 3 describes the data. Sections 4, 5, and 6 present the three layers in detail. Section 7 compares all models, and Section 8 concludes with limitations and directions for future work.

2. BACKGROUND

This section collects the definitions, formulas, and methods that the reader needs to follow the analysis. Throughout, ϕ denotes coefficients within the Gini equation (the trend slope and autoregressive terms), while β denotes coefficients on exogenous macroeconomic regressors.

We assume familiarity with basic linear algebra (matrix multiplication, systems of linear equations) and introductory probability (random variables, expectation, variance) at the level of a senior mathematics major.

2.1. The Gini Coefficient. The Gini coefficient is a measure of statistical dispersion that summarizes the inequality of a distribution. It is most commonly understood through the *Lorenz curve*, which plots the cumulative share of total income earned by the bottom $x\%$ of the population against x . If income were perfectly equal, the Lorenz curve would be the 45-degree line (everyone earns the same share); the more unequal the distribution, the more the curve bows away from this line.

Definition 2.1 (Gini Coefficient). Let A denote the area between the line of perfect equality and the Lorenz curve, and let B denote the area below the Lorenz curve. The *Gini coefficient* is

$$G = \frac{A}{A + B}.$$

Since the total area under the equality line is $1/2$ (it is a triangle with base and height both equal to 1), we have $A + B = 1/2$, so equivalently $G = 2A$. When everyone earns the same amount, the Lorenz curve coincides with the equality line, $A = 0$, and $G = 0$. When all income is concentrated in a single individual, A is as large as possible, and G approaches 1. The larger A relative to B , the more unequal the distribution (see Figure 1).

In practice, these annual Gini values are estimated from national household survey data; we use the published series directly.

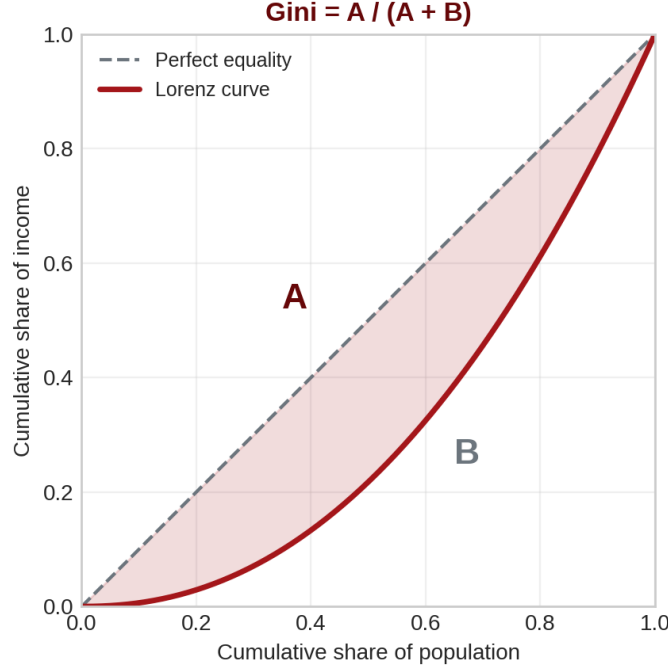


FIGURE 1. The Lorenz curve. Area A lies between the line of perfect equality and the curve; B lies below it, so $G = A/(A + B)$.

2.2. Ordinary Least Squares. Ordinary least squares (OLS) is the method we use to estimate the parameters in our models. The idea is simple: given a model that predicts values $\hat{y}_t$ from some inputs, find the parameter values that make the predictions as close as possible to the actual observed values y_t .

Specifically, OLS finds the parameters that minimize the total squared error:

$$\min \sum_t (y_t - \hat{y}_t)^2.$$

Squaring the errors means that big mistakes are penalized more heavily than small ones, which encourages the model to avoid big misses.

When the model is linear in its parameters (as all of our models are), the minimization has a closed-form solution. We arrange the predictor values into a matrix X and the observed values into a vector $\mathbf{y}$, and the OLS estimator is:

$$(1) \quad \hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y},$$

where $\hat{\beta}$ is the vector of estimated coefficients, X is the matrix of predictor values (one row per observation, one column per predictor), and $\mathbf{y}$ is the vector of values being predicted.

In our implementation, the OLS estimates are computed directly from the closed-form solution in Equation (1) using NumPy's linear algebra routines.

2.3. Maximum Likelihood Estimation. Starting from the ARMA model (Section 6.2), the models become too complex for OLS to handle directly. This is because the moving-average (MA) terms introduce past forecast errors into the prediction equation, and these errors depend on the parameters themselves—creating a circular dependency that OLS cannot resolve with a simple matrix formula.

In these cases, we use *conditional maximum likelihood estimation (MLE)*. The key idea is the same as OLS: find the parameter values that best fit the data. The difference is in how “best fit” is measured.

Definition 2.2 (Likelihood). The *likelihood* of a set of parameters is the probability of observing the data we actually observed, given those parameter values.

MLE finds the parameters that maximize this likelihood—or equivalently, that maximize the *log-likelihood* $\ln L$ (since the logarithm is an increasing function, maximizing $\ln L$ gives the same answer as maximizing L , but is computationally easier).

For models with normally distributed errors and variance σ^2 , the log-likelihood takes the form:

$$\ln \hat{L} = -\frac{n}{2}(\ln(2\pi\hat{\sigma}^2) + 1),$$

where n is the number of observations and $\hat{\sigma}^2$ is the estimated error variance. A higher (less negative) log-likelihood means the model fits the data more closely.

In practice, because the MA terms make the likelihood a complicated nonlinear function of the parameters, we cannot solve for the maximum by hand. Instead, we use *numerical optimization*: the computer systematically tries different parameter values, guided by an algorithm (in our case, the Nelder–Mead method from SciPy [11]), until it finds the combination that maximizes the log-likelihood.

It is important to note that OLS and MLE are doing the same job—finding the coefficients that best fit the data. OLS is simply the special case where a clean shortcut (the matrix formula (1)) exists. When the model is too complex for that shortcut, MLE provides a more general approach. We still use AIC and BIC for model selection (Section 2.4) regardless of whether the parameters were estimated by OLS or MLE.

2.4. Model Selection: AIC and BIC. When fitting a model, we must choose its structure: how many lags to include, which variables to use, and so on. A model with too few terms may miss important patterns (underfitting), while too many terms may fit noise rather than signal (overfitting). Information criteria provide a principled way to balance model fit against complexity.

Both the AIC and BIC are built on the maximized log-likelihood $\ln \hat{L}$, which measures how well a fitted model explains the observed data (Section 2.3). They combine this measure of fit with a penalty for the number of parameters k ; our implementation follows the standard formulations [13, 12]:

Definition 2.3 (Akaike Information Criterion).

$$(2) \quad \text{AIC} = 2k - 2 \ln \hat{L}.$$

Definition 2.4 (Bayesian Information Criterion).

$$(3) \quad \text{BIC} = k \ln n - 2 \ln \hat{L}.$$

In both formulas, the term $-2 \ln \hat{L}$ captures model fit (smaller is better), and the first term penalizes complexity. The BIC penalty $k \ln n$ grows with the sample size n and therefore penalizes additional parameters more heavily than the AIC penalty $2k$, especially in moderate-to-large samples.

In practice, we fit models across a grid of candidate structures, compute each criterion, and select the structure that yields the lowest score. We use BIC as our primary selection criterion

throughout this project because it is more conservative and better suited to our small sample ($n = 31$ training observations), consistent with standard practice in macroeconomic time-series forecasting [5].

Remark 2.1 (Comparing Information Criteria). Because models involving lagged variables or differencing (e.g., ARX, ARIMA) consume initial observations, the effective sample size n drops relative to a baseline AR(1) or trend model. As a result, the raw AIC and BIC values reported in Section 6 are computed over slightly different estimation windows than those in Section 5. While these criteria are robust for selecting the best specification *within* a specific model class, they should not be compared *across* model classes with different effective sample sizes.

2.5. Expanding-Window Rolling Forecast. To evaluate out-of-sample forecast performance without look-ahead bias, we use an *expanding-window rolling forecast* procedure [10]. The idea is to mimic the real-time situation of an economist who must forecast next year’s Gini value using only data available up to the present.

The procedure works as follows:

- (1) Fit the model on all data from 1963 to year $t - 1$.
- (2) Predict one step ahead: $\hat{y}_t$.
- (3) Reveal the actual value y_t and expand the training window.
- (4) Repeat for $t = 1994, 1995, \dots, 2023$.

This produces 30 one-step-ahead forecasts. Because the model is re-estimated at each step using all available data, the parameter estimates gradually incorporate new information as the training window expands. At no point does the model “see” future data before making its prediction.

2.6. Forecast Evaluation Metrics. We assess forecast accuracy using three standard metrics. Let y_t and $\hat{y}_t$ denote the actual and predicted values, and let $e_t = y_t - \hat{y}_t$ denote the forecast error.

Definition 2.5 (Mean Absolute Error).

$$(4) \quad \text{MAE} = \frac{1}{n} \sum_{t=1}^n |e_t|.$$

MAE measures the average magnitude of forecast errors in the same units as the original series.

Definition 2.6 (Root Mean Squared Error).

$$(5) \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}.$$

RMSE places greater weight on large errors than MAE due to the squaring.

Definition 2.7 (Mean Absolute Percentage Error).

$$(6) \quad \text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{e_t}{y_t} \right|.$$

MAPE expresses errors as a percentage of the actual values. A MAPE under 5% is generally considered indicative of a highly accurate forecast.

3. DATA

Our primary dataset consists of 61 annual observations of the U.S. Gini coefficient from 1963 to 2023, obtained from Our World in Data, which compiles the series from the World Bank’s Poverty and Inequality Platform [1]. Over this period, the Gini coefficient rose from approximately 0.37 to approximately 0.42, reflecting a persistent upward trend in income inequality. Figure 2 displays the full time series.

We split the data into a training period (1963–1993, 31 years) and a test period (1994–2023, 30 years). All model fitting and parameter selection are performed on the training data, and all forecast evaluations are conducted on the test data via the rolling procedure described in Section 2.5.

For Layer 3, we also use three macroeconomic time series from the Federal Reserve Economic Data (FRED) database [2]: GDP (Gross Domestic Product), used as year-over-year change ΔGDP_t ; CPI (Consumer Price Index), used as year-over-year change ΔCPI_t to proxy inflation; and the unemployment rate, used as year-over-year change ΔUNRATE_t . Each variable is tested at lags 1, 2, and 3, yielding 9 candidate exogenous regressors.

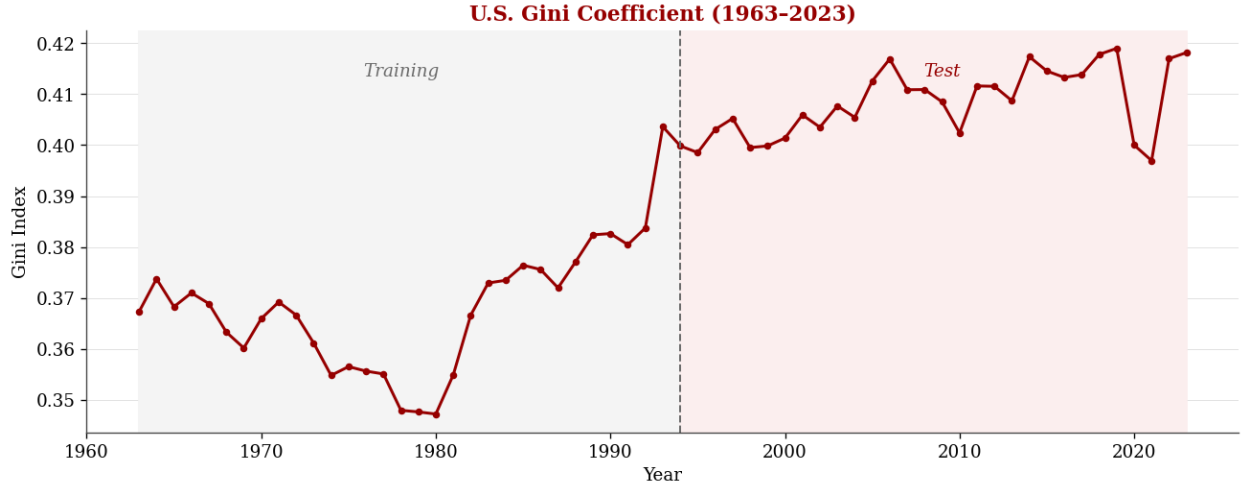


FIGURE 2. U.S. Gini coefficient, 1963–2023, with the 1994 train/test split.

4. LAYER 1: LINEAR TREND

4.1. Model. Layer 1 establishes a simple baseline by fitting a deterministic linear time trend to the Gini data:

$$(7) \quad y_t = c + \phi_1(t - t_0) + \varepsilon_t,$$

where y_t is the Gini coefficient in year t , c is the baseline level at the starting year $t_0 = 1963$, ϕ_1 is the average annual change (slope), and ε_t is the error. The parameters (c, ϕ_1) are estimated via OLS and re-estimated each year using the expanding-window rolling forecast procedure.

4.2. **Worked Example: Predicting y_{1994} .** Using the training window 1963–1993 ($n = 31$ years), OLS yields $\hat{c} = 0.3580$ and $\hat{\phi}_1 = 0.000654$. The prediction for 1994 is:

$$\hat{y}_{1994} = 0.3580 + 0.000654 \times (1994 - 1963) = 0.3580 + 0.0203 = 0.3783.$$

The actual value is $y_{1994} = 0.3999$, giving an error of $+0.0216$. The training window then expands to 1963–1994, OLS is refit, and the process repeats for 1995.

4.3. **Results.** Over the 30-year test period, Layer 1 achieves:

Metric	Value
MAE	0.0081
RMSE	0.0106
MAPE	2.01%

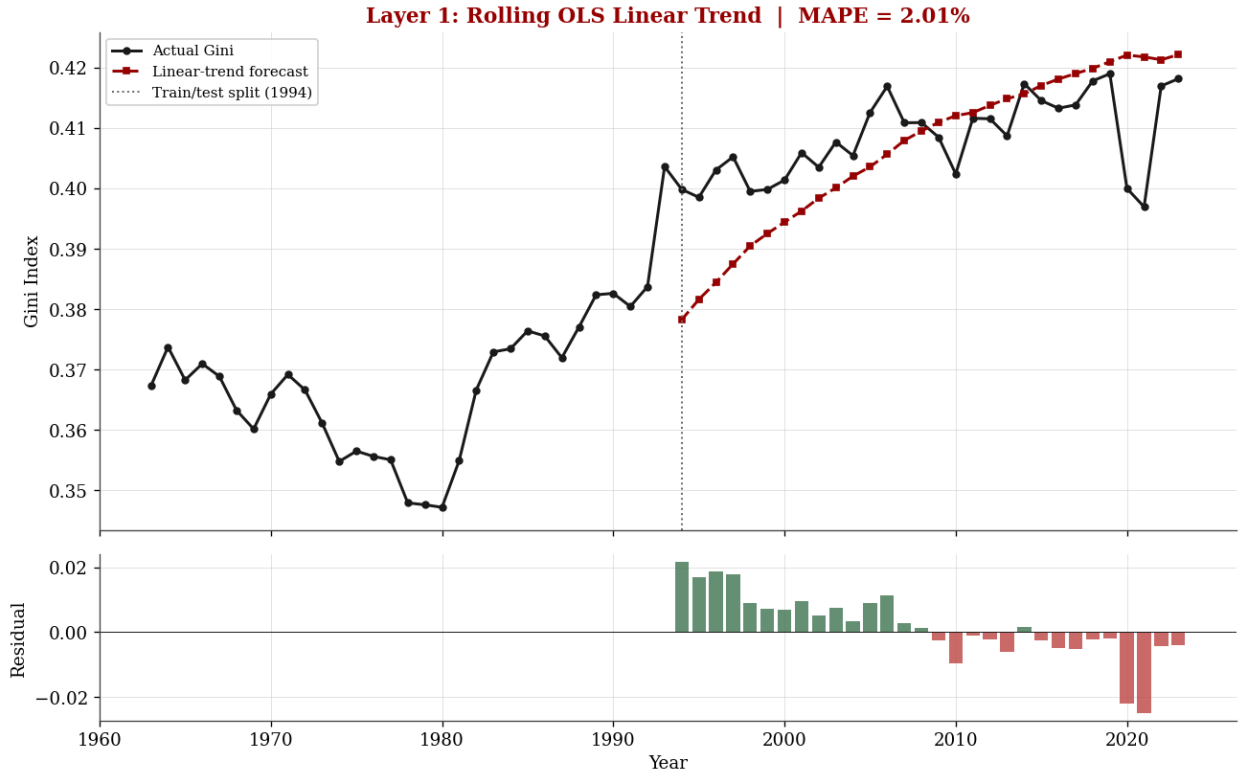


FIGURE 3. Layer 1 rolling linear-trend forecast vs. actual Gini, with residuals. The persistent residual pattern motivates the AR model.

A MAPE of 2.01% is reasonable, but as shown in Figure 3, the model has systematic problems: the straight line consistently over-predicts during the 1990s–2000s, and the residuals show persistent patterns rather than random scatter. When the model misses high one year, it tends to miss high the next year too. A straight line keeps the same slope even when the data changes direction, and it cannot adapt.

These patterned errors suggest that a model which can learn from its own past values should do better—motivating the autoregressive model in Layer 2.

5. LAYER 2: AUTOREGRESSIVE MODEL

5.1. The AR(p) Model. An autoregressive model of order p , denoted $\text{AR}(p)$, expresses the current value of a time series as a linear combination of its p most recent past values plus a random disturbance [8].

Definition 5.1 (AR(p) Model). A time series $\{y_t\}$ follows an $\text{AR}(p)$ process if

$$(8) \quad y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

where:

- y_t is the value of the series at time t (the Gini coefficient in year t),
- c is an intercept (constant term),
- $\phi_1, \dots, \phi_p$ are the autoregressive coefficients,
- ε_t is a random error term representing unpredictable shocks, assumed to have mean zero and constant variance σ^2 .

The key difference from Layer 1 is the predictor: instead of using a time index $(t - t_0)$, the AR model uses the series' own past values. The coefficients ϕ_i capture the *persistence* of the series—how strongly past values influence the present. A coefficient ϕ_1 close to 1 in an $\text{AR}(1)$ model indicates that the series changes slowly, with shocks dissipating only gradually over time.

5.2. Order Selection. Using the training data from 1963 to 1993, we fit $\text{AR}(p)$ models for $p = 1, 2, \dots, 10$ and computed both the AIC and BIC for each. The top three results are shown in Table 1.

TABLE 1. Information criteria for $\text{AR}(p)$ models on the 1963–1993 training sample.

p	AIC	BIC
1	−218.93	−214.72
2	−211.66	−206.19
3	−205.69	−199.03

Both criteria are minimized (most negative) at $p = 1$. Every additional lag makes the score worse: adding more history does not help, because last year's value alone is enough to predict this year's. Both criteria agree: $p = 1$. Figure 4 illustrates the dramatic improvement of this $\text{AR}(1)$ model over the baseline trend.

5.3. Worked Example: Predicting y_{1994} . With $p = 1$ selected, the $\text{AR}(1)$ model is:

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t.$$

Step 1. Training window: 1963–1993 ($n = 31$ years).

Step 2. Fit $\text{AR}(1)$ via OLS—find $\hat{c}$ and $\hat{\phi}_1$ that minimize:

$$\min_{c, \phi_1} \sum_{t=1964}^{1993} (y_t - [c + \phi_1 y_{t-1}])^2.$$

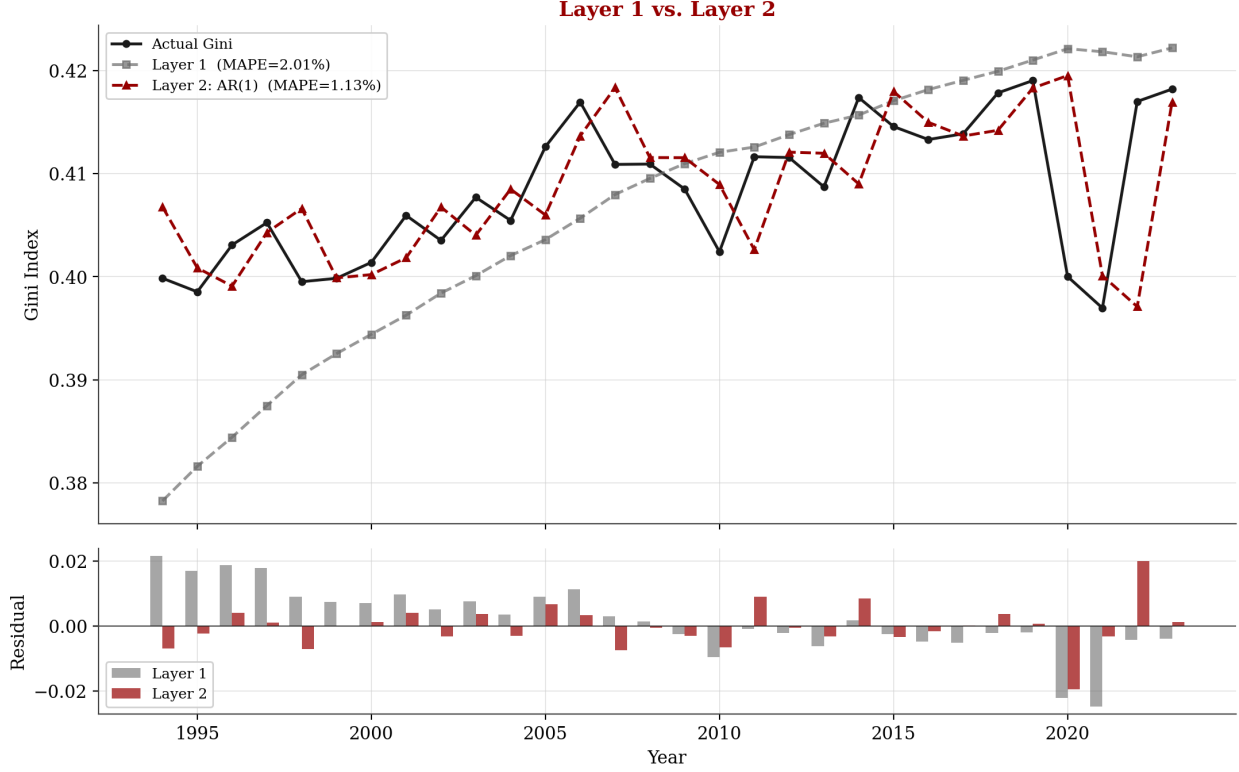


FIGURE 4. Layer 1 vs. Layer 2. The AR(1) forecast (maroon) tracks the data far more closely than the trend line.

The key difference from Layer 1: the predictor is now last year's Gini, not a time index. The initial training fit yields $\hat{c} = -0.0180$ and $\hat{\phi}_1 = 1.0524$.¹

Step 3. Predict (using $y_{1993} = 0.4036$):

$$\hat{y}_{1994} = \underbrace{-0.0180}_{\hat{c}} + \underbrace{1.0524}_{\hat{\phi}_1} \times \underbrace{0.4036}_{y_{1993}} = -0.0180 + 0.4248 = 0.4068.$$

Step 4. Compare: actual $y_{1994} = 0.3999$, error = -0.0069 . Layer 1's error for the same year was $+0.0216$ —more than three times larger.

Step 5. Expand the training window to 1963–1994, refit OLS, predict 1995, and repeat.

5.4. Results. Over the 30-year test period, the final rolling-fit parameter estimates (after incorporating all data through 2022) are $\hat{c} = 0.01230$ and $\hat{\phi}_1 = 0.97038$. The persistence parameter $\hat{\phi}_1 \approx 0.97$ means that approximately 97% of last year's Gini value carries forward into this year, confirming the slow-moving nature of inequality. Comparable persistence appears in GDP series, which are routinely modeled with ARIMA baselines; against these baselines, more elaborate hybrid models deliver only uneven gains [4].

Table 2 compares Layer 2 with Layer 1.

¹An estimate $\hat{\phi}_1 > 1$ implies a (mildly) explosive process and is a small-sample artifact of the short 31-year initial window. As the rolling window expands, the estimate settles to the stationary value $\hat{\phi}_1 \approx 0.97$ reported below.

TABLE 2. Forecast evaluation: Layer 1 vs. Layer 2.

Model	MAE	RMSE	MAPE	Improvement
Layer 1 (trend)	0.0081	0.0106	2.01%	—
Layer 2 (AR(1))	0.0046	0.0066	1.13%	$\approx 44\%$

Layer 2 cuts the MAPE nearly in half. The AR(1) model tracks the actual data much more closely than the trend line, and its residuals are smaller and more balanced. One of the largest misses occurred in 2020, when the Gini dropped sharply to 0.400 (likely reflecting pandemic-era transfer payments and income measurement changes) while the model predicted 0.420 based on 2019’s higher value.

5.5. Limitations. Despite its strong performance, the AR(1) model has clear limitations that motivate Layer 3:

- **Only looks at itself:** AR(1) uses past Gini values alone—it has no idea what the economy is doing.
- **Very slow to adjust:** with $\phi \approx 0.97$, a shock takes roughly 23 years to half-fade, which is long relative to the 31-year initial training window—barely more than one half-life of data.
- **No self-correction:** if the model over-predicts one year, it has no mechanism to “learn from that mistake” and pull the next prediction back down.

We need a model that can use economic data, correct its own errors, and handle trending data—this is what Layer 3 addresses.

6. LAYER 3: EXTENDED MODELS

Layer 3 tests four extensions of the AR(1) model, each adding one capability. Table 3 summarizes the differences.

TABLE 3. Layer 3 model capabilities.

Model	Past Gini	Past errors	Differenced	Macro data
AR(1)	✓			
ARX	✓			✓
ARMA	✓	✓		
ARIMA	✓	✓	✓	
ARIMAX	✓	✓	✓	✓

6.1. ARX: Adding Macroeconomic Variables. The “X” in ARX stands for *exogenous*: an outside variable not derived from the Gini series itself. The ARX model adds macroeconomic regressors to the AR(1) equation:

$$y_t = c + \phi_1 y_{t-1} + \beta_1 X_{1,t-\ell_1} + \beta_2 X_{2,t-\ell_2} + \cdots + \varepsilon_t,$$

where each $X_{k,t-\ell_k}$ is a lagged macroeconomic variable and β_k is its coefficient.

We test three macroeconomic variables from FRED— ΔGDP , ΔCPI , and ΔUNRATE —each at lags 1, 2, and 3, yielding 9 candidate regressors. We perform a BIC subset search

over all $2^9 = 512$ combinations to find the best specification. Because the model is still linear and contains no MA terms, estimation uses OLS.

BIC selection result: $\text{AR}(1) + \Delta\text{GDP}_{t-2} + \Delta\text{CPI}_{t-1}$ (BIC = -199.91).

Selected equation:

$$y_t = c + \phi_1 y_{t-1} + \beta_1 \Delta\text{GDP}_{t-2} + \beta_2 \Delta\text{CPI}_{t-1} + \varepsilon_t.$$

Rolling forecast results: MAE = 0.0048, RMSE = 0.0059, MAPE = 1.17%.

Adding macroeconomic variables does not improve forecasts beyond lagged Gini alone. The selected ΔGDP_{t-2} term survives the BIC penalty only because it trims the in-sample residual sum of squares by just enough to offset the $\ln n$ cost of one extra parameter; its estimated coefficient is nonetheless economically negligible ($\hat{\beta}_1 \approx -0.00004$). The ΔCPI_{t-1} coefficient is somewhat larger but likewise too small to matter, and neither term improves out-of-sample accuracy.

6.2. ARMA: Learning from Past Errors. The “MA” in ARMA stands for *moving average*—not of the data itself, but of past forecast errors. An ARMA model combines the autoregressive component (past values of the series) with a moving-average component (past prediction mistakes) [9]:

$$(9) \quad y_t = c + \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR: past values}} + \underbrace{\theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}}_{\text{MA: past errors}} + \varepsilon_t,$$

where ε_{t-j} is the forecast error from j periods ago and θ_j is its coefficient. The MA terms allow the model to “learn from its own mistakes”: if it over-predicted last year, the MA correction can pull the next prediction down.

Because the error terms ε_{t-j} depend on the parameters themselves, OLS cannot be used. Instead, we estimate the parameters using conditional maximum likelihood estimation (Section 2.3).

We search over all $\text{ARMA}(p, q)$ combinations with $p, q \in \{0, 1, 2, 3\}$ and select the specification with the lowest BIC.

BIC selection result: $\text{ARMA}(2, 2)$ (BIC = -238.46).

Selected equation:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t.$$

Rolling forecast results: MAE = 0.0053, RMSE = 0.0074, MAPE = 1.31%.

Error correction adds complexity but does not improve out-of-sample accuracy. The MA terms overfit to training-period patterns that do not generalize to the test period.

6.3. ARIMA: Handling Trending Data. The “I” in ARIMA stands for *integrated*, referring to the operation of *differencing* the series before modeling. Instead of modeling y_t directly (which trends upward), we model the year-over-year change $\Delta y_t = y_t - y_{t-1}$, which is stationary (its statistical properties do not change over time).

An $\text{ARIMA}(p, d, q)$ model differences the series d times, then fits an $\text{ARMA}(p, q)$ model on the differenced data [9]:

$$(10) \quad \Delta y_t = c + \phi_1 \Delta y_{t-1} + \cdots + \phi_p \Delta y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t,$$

where c is the constant (drift) term. To produce a forecast of y_t (in levels), we predict Δy_t using the model and then convert back: $\hat{y}_t = y_{t-1} + \widehat{\Delta y}_t$.

We fix $d = 1$ due to the size of our sample and search over $p, q \in \{0, 1, 2, 3\}$.

BIC selection result: ARIMA(3,1,3) (BIC = -210.25).

Selected equation:

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \phi_2 \Delta y_{t-2} + \phi_3 \Delta y_{t-3} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + \varepsilon_t.$$

Rolling forecast results: MAE = 0.0057, RMSE = 0.0082, MAPE = 1.40%.

Differencing plus MA terms does not beat AR(1). The high-order model (7 parameters) overfits the small 31-year training set.

6.4. ARIMAX: Combining Everything. ARIMAX combines all three extensions: differencing, MA error correction, and macroeconomic regressors. It takes the ARIMA model and adds exogenous variables:

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \cdots + \phi_p \Delta y_{t-p} + \beta_k \Delta X_{k,t-\ell_k} + \cdots + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

Using ARIMA(3,1,3) as the baseline, we perform a BIC subset search over all macroeconomic variable combinations.

BIC selection result: No exogenous variables improve the BIC. ARIMAX collapses to ARIMA(3,1,3).

This is a meaningful null result: even with the most flexible model structure available, macroeconomic data does not provide additional forecasting power beyond what the Gini series' own history already captures. The Gini coefficient's year-to-year persistence is so dominant that external economic indicators are redundant for short-term prediction.

Rolling forecast results: Identical to ARIMA—MAE = 0.0057, RMSE = 0.0082, MAPE = 1.40% (see Figure 5 for a visual comparison of the Layer 3 models).

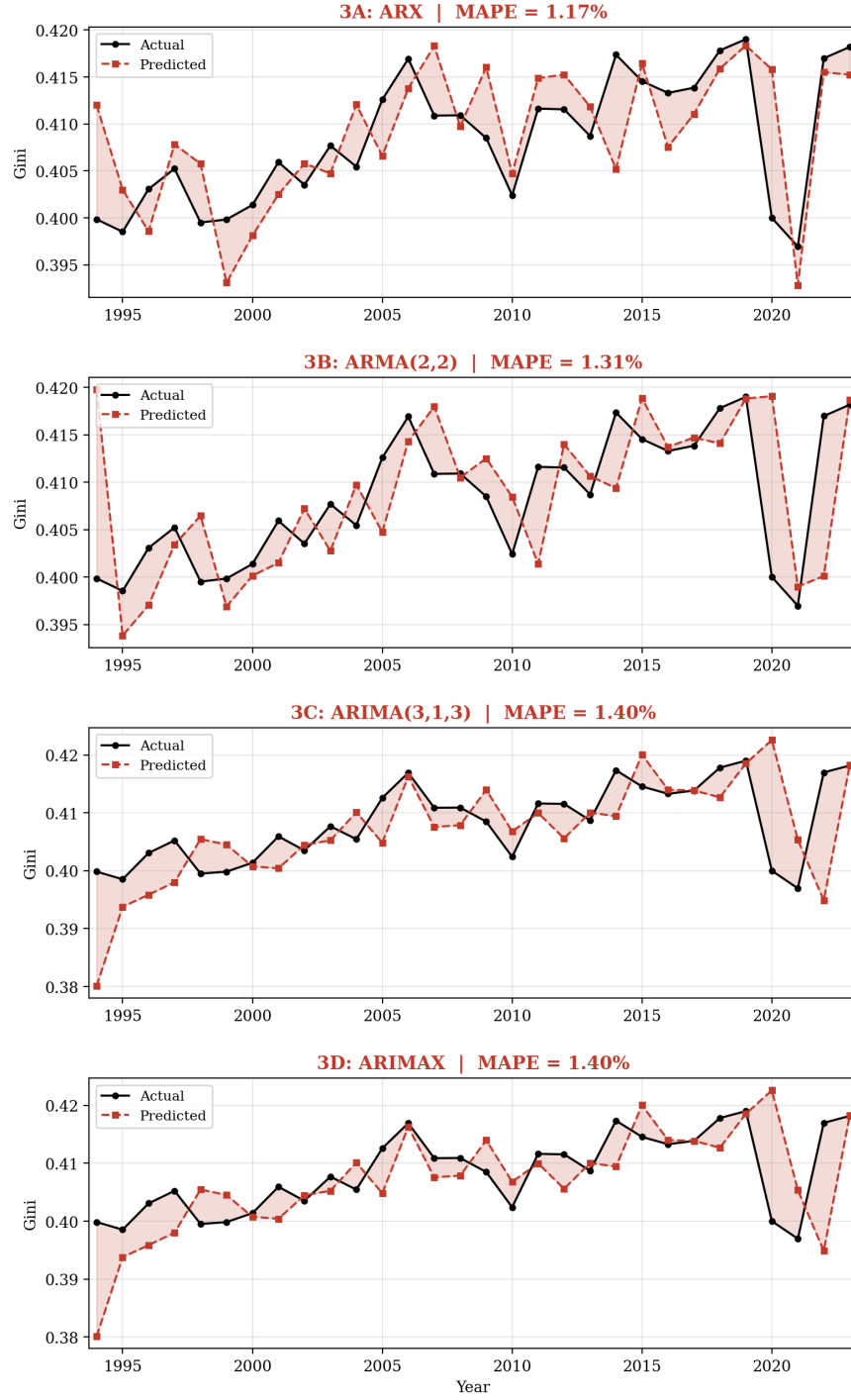


FIGURE 5. Layer 3 models: forecast vs. actual.

7. RESULTS

Table 4 compares all six models across the three error metrics on the 30-year test period.

TABLE 4. Forecast evaluation: all models, 1994–2023.

Model	MAE	RMSE	MAPE (%)
Layer 1: Linear Trend	0.0081	0.0106	2.01
Layer 2: AR(1)	0.0046	0.0066	1.13
3A: ARX	0.0048	0.0059	1.17
3B: ARMA(2,2)	0.0053	0.0074	1.31
3C: ARIMA(3,1,3)	0.0057	0.0082	1.40
3D: ARIMAX	0.0057	0.0082	1.40

The central finding is that the simple AR(1) model is the best overall forecaster. It achieves the lowest MAE and MAPE of any model, and no extension improves on it across all three metrics simultaneously. Adding complexity—whether through macroeconomic covariates (ARX), error correction terms (ARMA), differencing (ARIMA), or all three combined (ARIMAX)—does not yield a consistent improvement. The one partial exception is ARX, which edges out AR(1) on RMSE (0.0059 vs. 0.0066) while remaining worse on MAE and MAPE; since the gain shows up on only one metric and the selected ΔGDP and ΔCPI coefficients are economically negligible, we do not treat it as a meaningful improvement.

This result has a clear economic interpretation: U.S. income inequality, as measured by the Gini coefficient, is dominated by its own persistence. The estimated AR(1) coefficient $\hat{\phi}_1 \approx 0.97$ means that knowing last year’s Gini value is, by itself, an extremely strong predictor of this year’s value. Macroeconomic fluctuations in GDP, inflation, and unemployment operate on shorter time scales and have effects too small to improve year-ahead inequality forecasts. ARIMA-based time-series models are an established tool for forecasting Gini and inequality series [3, 7]; our results suggest that for one-step-ahead U.S. forecasts, a simple AR(1) performs at least as well.

All models share a common weakness: they struggle with sudden, unpredictable shocks. The 2020 COVID-19 pandemic caused a sharp drop in measured inequality (likely due to large-scale government transfer payments), producing one of the largest forecast errors for every model (Figure 6). The 2022 rebound, when inequality snapped back to pre-pandemic levels, also generated large errors for the more complex models. These structural shocks are, by their nature, outside the scope of any model trained on historical patterns.

8. CONCLUSION

This paper has presented a layered modeling framework for forecasting U.S. income inequality as measured by the Gini coefficient. Layer 1 established a deterministic linear trend baseline with a MAPE of 2.01%. Layer 2 introduced an AR(1) model that cut the error nearly in half (MAPE 1.13%) by exploiting the strong year-to-year persistence in inequality. Layer 3 tested four extensions—ARX, ARMA, ARIMA, and ARIMAX—none of which improved upon the AR(1) baseline.

The main conclusions are:

- (1) The Gini coefficient is an extremely persistent series ($\hat{\phi}_1 \approx 0.97$), meaning that last year’s value alone is a powerful predictor of this year’s.
- (2) Macroeconomic covariates (GDP growth, inflation, unemployment) do not significantly improve one-step-ahead Gini forecasts. This is a legitimate, reportable null

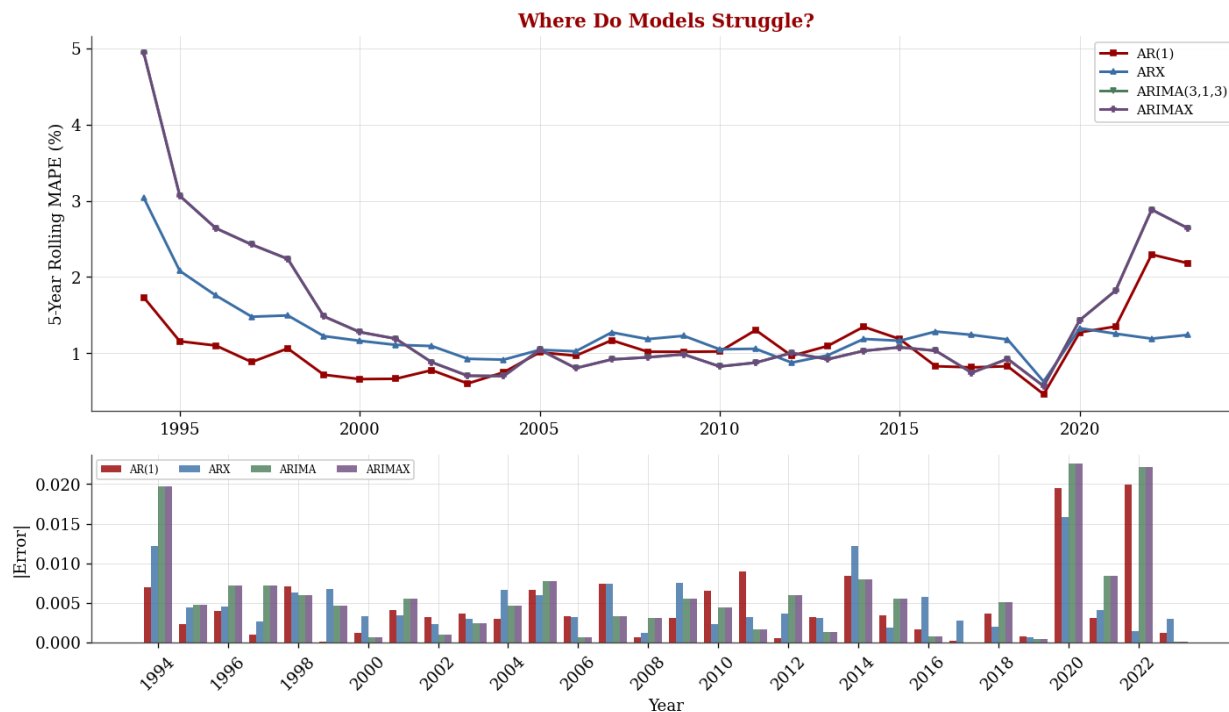


FIGURE 6. Rolling MAPE and per-year absolute error. All models spike at the 2020 shocks.

result: the Gini's own persistence already absorbs the slow-moving signal, leaving little for short-term macroeconomic fluctuations to add.

- (3) Adding model complexity (MA terms, differencing, exogenous regressors) leads to overfitting on a small training sample, worsening out-of-sample performance.

8.1. Limitations. Several limitations should be noted. First, with only 61 annual observations, the sample is small for complex time-series models—the ARIMA(3,1,3) model, for example, estimates 7 parameters from 31 training observations. Second, the Gini coefficient compresses an entire income distribution into a single number, missing important features such as top-1% income concentration, the distinction between wealth and income inequality, and regional variation. Third, all models assume a stable underlying process over 60 years, during which the U.S. economy underwent substantial structural changes.

8.2. Future Directions. Several extensions could strengthen the analysis: alternative inequality measures such as the Palma ratio or top income shares; prediction intervals via bootstrap methods to quantify forecast uncertainty; regime-switching models that allow parameters to change across economic eras; structural policy variables such as tax rates, educational attainment, and technology shocks; and higher-frequency data (quarterly or monthly inequality proxies) to increase sample size and capture faster-moving dynamics.

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